

Ice Cream Parlour Management System (ICPMS)

Version

1.0

Project Type

Production-Grade Business Management Software

Technology Stack

Frontend:

- React.js
- Tailwind CSS
- React Router
- Axios
- Recharts

Backend:

- Node.js
- Express.js
- JWT Authentication
- REST APIs

Database:

- MongoDB
- Mongoose ODM

Storage:

- Cloudinary (Product Images)

Deployment:

- Frontend: Vercel
 - Backend: Render / Railway
 - Database: MongoDB Atlas
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1. Project Overview

The Ice Cream Parlour Management System (ICPMS) is a full-scale business management application designed to automate and streamline daily operations of an Ice Cream Parlour.

The system will provide tools for sales management, billing, inventory tracking, customer management, employee management, analytics, discounts, and role-based access control.

Unlike an academic project, this software is intended to simulate a real-world product that can be deployed and used by actual business owners.

2. Business Problem

Many small and medium-sized ice cream parlours still manage their operations manually using notebooks, spreadsheets, or basic billing systems.

This creates several challenges:

- Inaccurate inventory tracking
- Difficult sales monitoring
- No centralized customer records
- Manual billing errors
- Poor employee accountability
- Lack of business insights
- No real-time stock visibility

The proposed system aims to solve these issues through digital automation.

3. Project Goals

The primary goals of the project are:

- Automate daily business operations
- Simplify billing and invoice generation
- Track inventory in real-time
- Manage products efficiently
- Maintain customer records
- Generate sales reports
- Monitor employee activities
- Improve decision-making through analytics
- Reduce manual errors
- Increase operational efficiency

4. Target Users

Admin

Business owner with complete system access.

Responsibilities:

- Manage users
- Manage inventory
- View reports
- Configure system settings
- Manage employees
- Create offers

Manager

Responsible for store operations.

Responsibilities:

- Monitor sales
- Track inventory
- Manage products
- View reports
- Manage staff

Cashier

Handles customer billing.

Responsibilities:

- Create orders
 - Generate invoices
 - Accept payments
 - Manage customer records
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Inventory Staff

Manages stock operations.

Responsibilities:

- Stock updates
 - Inventory monitoring
 - Low stock handling
 - Supplier coordination
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5. Core Business Modules

Module 1: Authentication & Authorization

Features:

- Secure Login
 - Logout
 - JWT Authentication
 - Protected Routes
 - Role-Based Access Control
 - Password Change
 - Session Management
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Module 2: Dashboard

Features:

- Today's Sales
 - Monthly Revenue
 - Orders Count
 - Customer Count
 - Low Stock Alerts
 - Revenue Trends
 - Top Selling Products
 - Recent Orders
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Module 3: Product Management

Features:

- Add Product
- Update Product
- Delete Product
- Product Categories
- Product Images
- Product Status
- Product Search

Product Examples:

- Chocolate Cone
 - Vanilla Cup
 - Mango Sundae
 - Strawberry Shake
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Module 4: Billing & Invoice System

Features:

- Create Bill
- Add Products to Cart
- Tax Calculation
- Discount Calculation
- Payment Processing
- Invoice Generation
- Invoice Printing
- PDF Invoice Download

Payment Methods:

- UPI
 - Cash
 - Card
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Module 5: Customer Management

Features:

- Customer Registration
- Customer Search

- Mobile Number Tracking
 - Purchase History
 - Loyalty Points
 - Customer Analytics
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Module 6: Inventory Management

Features:

- Stock Tracking
- Stock Updates
- Low Stock Alerts
- Stock Consumption Tracking
- Inventory History
- Inventory Reports

Inventory Examples:

- Milk
 - Chocolate Syrup
 - Cones
 - Cups
 - Toppings
 - Dry Fruits
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Module 7: Order Management

Features:

- New Orders
- Order Status Tracking
- Completed Orders
- Cancelled Orders
- Order History

Order Status:

- Pending
 - Preparing
 - Ready
 - Completed
 - Cancelled
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Module 8: Employee Management

Features:

- Employee Records
 - Attendance Tracking
 - Role Assignment
 - Shift Tracking
 - Salary Information
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Module 9: Offers & Discounts

Features:

- Percentage Discount
 - Flat Discount
 - Seasonal Offers
 - Festival Offers
 - Combo Offers
 - Happy Hour Offers
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Module 10: Reports & Analytics

Features:

- Daily Sales Report
 - Weekly Sales Report
 - Monthly Sales Report
 - Product Performance Report
 - Customer Analytics
 - Inventory Report
 - Revenue Analytics
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6. Functional Requirements

The system shall:

- Allow authorized users to log in.
- Allow product creation and updates.
- Allow billing and invoice generation.
- Allow inventory management.

- Allow customer record management.
 - Allow employee management.
 - Allow discount management.
 - Generate business reports.
 - Store product images using Cloudinary.
 - Maintain complete sales history.
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7. Non-Functional Requirements

Performance:

- Dashboard loads within 3 seconds.

Security:

- JWT Authentication
- Password Hashing using bcrypt
- Role-Based Access

Scalability:

- Support future expansion to multiple branches.

Reliability:

- Data consistency and validation.

Usability:

- Mobile-friendly responsive interface.

Maintainability:

- Modular code structure.
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8. Cloudinary Integration

Purpose:

Store and manage product images.

Features:

- Image Upload

- Image Update
- Image Delete
- Optimized Image Delivery
- Secure Storage

Supported Image Types:

- JPG
 - PNG
 - WEBP
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9. Future Scope

Potential future enhancements:

- Multi-Branch Management
 - Online Ordering
 - Mobile Application
 - QR Menu System
 - Loyalty Rewards Program
 - AI-Based Sales Forecasting
 - WhatsApp Invoice Sharing
 - Customer Feedback Module
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10. Project Deliverables

The final system shall include:

- Professional Dashboard
- Authentication System
- Product Management
- Inventory Management
- Billing System
- Customer Management
- Employee Management
- Sales Analytics
- Reports Module
- Responsive UI
- Cloudinary Integration
- MongoDB Database
- Complete REST API Backend
- Production-Ready Source Code

End of Document